

IKERASSASSUAQ PRINS CHRISTIAN, Greenland

WMO: 043900

Lat: 60.055N

Lon: 43.165W

Elev: 19

StdP: 101.10

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.5	-9.3	-23.8	0.4	-7.8	-21.7	0.5	-7.5	29.8	-0.9	28.2	-1.5	10.8	270	0.969

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	14.2	8.7	12.6	7.9	11.2	7.2	9.9	12.8	8.9	11.1	8.1	10.0	5.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	6.9	9.9	7.6	6.5	9.1	6.8	6.2	8.5	29.1	13.1	26.8	11.3	25.1	10.0	15.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	21.2	18.4	DB	-12.1	17.6	2.5	2.2	-13.9	19.2	-15.3	20.5	-16.7	21.7	-18.5	23.2	(4)
(5)			WB	-13.4	11.6	2.1	1.8	-14.9	12.8	-16.1	13.9	-17.3	14.9	-18.8	16.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-3.2	-3.6	-3.2	-0.1	2.5	5.0	7.2	7.6	5.6	2.3	-0.2	-2.1	
	DBStd	4.66	2.79	3.01	2.96	2.19	1.96	2.21	2.25	2.26	2.18	2.29	2.24	2.68	
	HDD10.0	3112	410	382	409	303	233	151	92	79	134	237	307	375	
	HDD18.3	6141	668	615	668	553	491	400	346	331	383	496	557	633	
	CDD10.0	13	0	0	0	0	0	1	4	7	1	0	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	6.8	8.7	9.2	8.5	7.5	6.6	5.6	4.8	4.5	5.7	6.3	6.9	7.7	
Precipitation	PrecAvg	2098	222	171	138	156	133	148	128	174	214	167	179	209	
	PrecMax	3289	514	494	314	421	338	457	322	548	505	709	547	674	
	PrecMin	1097	55	6	6	24	16	7	25	17	46	23	54	56	
	PrecStd	671	110	123	82	99	81	92	75	106	110	137	118	140	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	3.1	4.5	8.3	11.5	14.2	17.2	17.6	13.7	10.7	7.3	8.0	
		MCWB	2.4	2.3	2.8	4.8	5.7	8.7	11.0	10.7	7.2	4.8	3.9	3.4	
	2%	DB	2.4	2.3	2.9	5.4	8.8	11.7	14.3	14.3	11.6	7.9	5.0	4.0	
		MCWB	1.2	1.7	1.5	2.4	4.3	6.9	9.1	8.9	6.2	4.0	2.5	2.5	
	5%	DB	1.4	1.3	1.7	4.0	6.9	10.0	12.2	12.5	10.2	6.5	3.7	2.6	
		MCWB	0.4	0.2	0.5	1.5	3.6	6.0	7.9	8.2	6.0	3.2	1.4	1.0	
	10%	DB	0.4	0.2	0.7	2.9	5.4	8.5	10.8	11.1	8.8	5.4	2.8	1.5	
		MCWB	-0.4	-1.0	-0.6	1.0	3.0	5.5	7.3	7.6	5.3	2.6	0.6	0.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.7	3.8	5.1	6.7	9.8	11.6	11.7	9.5	6.7	5.5	4.5
			MCDB	3.6	2.9	4.1	7.6	9.6	12.9	15.5	15.7	11.1	8.6	6.0	6.0
2%		WB	1.6	1.9	1.9	3.2	5.1	7.7	9.7	10.2	7.9	5.2	3.2	2.6	
		MCDB	2.0	2.2	2.4	4.5	7.3	10.0	13.0	12.9	9.7	6.7	4.2	3.7	
5%		WB	0.6	0.5	0.8	2.2	4.2	6.7	8.7	9.0	6.7	4.2	2.1	1.4	
		MCDB	1.1	0.9	1.4	3.2	5.8	8.8	11.0	11.2	8.7	5.6	3.1	2.5	
10%		WB	-0.2	-0.6	-0.2	1.5	3.5	6.0	7.9	8.2	6.1	3.3	1.2	0.4	
		MCDB	0.3	-0.1	0.5	2.4	4.9	7.9	9.9	10.1	7.9	4.8	2.3	1.2	
Mean Daily Temperature Range	5% DB	MDBR	3.0	3.2	3.3	3.4	3.8	4.7	4.9	4.4	3.8	3.2	2.9	2.9	
		MCDBR	3.6	3.6	3.8	5.1	6.7	7.5	7.7	7.1	6.0	4.7	4.0	3.7	
		MCWBR	3.2	3.4	3.2	3.4	4.0	4.3	4.1	3.7	3.5	3.1	3.3	3.1	
	5% WB	MCDBR	3.5	3.4	3.4	4.4	5.3	6.4	6.4	5.7	4.8	4.0	3.6	3.5	
		MCWBR	3.5	3.5	3.2	3.4	3.6	4.1	4.0	3.6	3.4	3.2	3.4	3.2	
Clear-Sky Solar Irradiance	taub	0.272	0.289	0.306	0.349	0.352	0.356	0.356	0.363	0.339	0.293	0.250	0.226		
	taud	2.065	2.254	2.293	2.189	2.248	2.313	2.382	2.362	2.364	2.347	2.162	1.980		
	Ebn at Noon	471	715	827	844	871	873	865	826	781	704	515	372		
	Edh at Noon	46	67	89	119	121	116	106	100	84	60	40	31		
All-Sky Solar Radiation	RadAvg	0.30	0.92	2.33	4.03	5.25	5.28	5.01	3.89	2.60	1.40	0.47	0.19		
	RadStd	0.04	0.11	0.19	0.37	0.57	0.49	0.28	0.33	0.28	0.15	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.87	+2.90	N/A	+0.61	N/A	-0.63	-272	-258	N/A	N/A

Nomenclature: See separate page